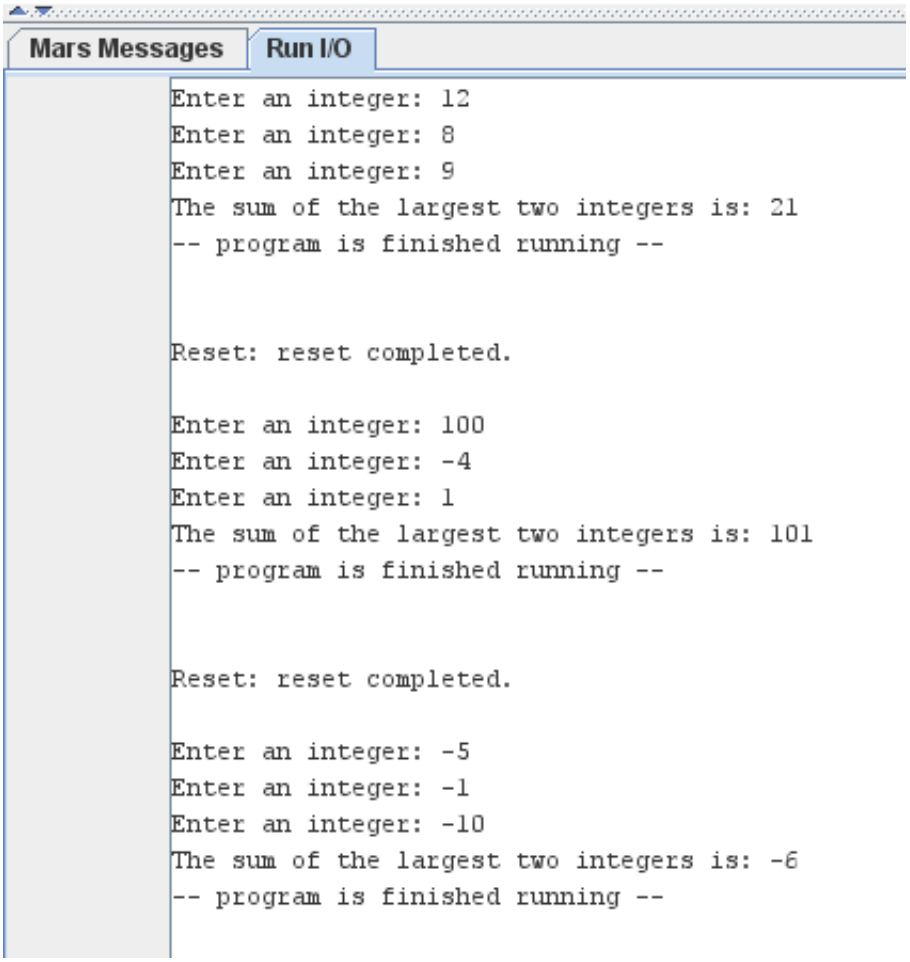
Lab4-2 CO Report

1. Austin Sternberg
2. lab4-2.asm
3. This program reads in three integers, then sums the largest of the two. Lines 25-48 prompt the user for integers and read their value in three separate times, storing their values in $t0, $t1, $t2. Lines 55 & 56 initialize our largest and second largest registers to the values of the first and second integers. From there, lines 59-61 start by checking if the first value is larger or equal to the second, in which case it will jump to check_ab. Otherwise, it will flip the values in the largest and second largest spots ($t4 & $t5). The check_ab section starts by seeing if the third value is greater than or equal to the largest value, in which case it will jump to check_ac. If not, the next line checks if c is greater than or equal to the second largest value, in which case it will jump to check_bc. If neither are true, the program will jump to the end. The check_ac section moves the value in the largest place to the second largest place and then sets the largest value to the value of the third integer and finally jumps to the end. The check_bc section replaces the value of the second largest with the value of the third integer. The end_check section adds the values of the largest and second largest values (stored in registers $t4 & $t5). Then this value is printed out and the program exits.

4. 

5. Unlike the first part of this assignment, this one took me a bit longer to figure out. While I had gotten it to work under certain conditions, it wasn't fully correct as switching the order of the input would lead to incorrect results. Eventually, I figured out this solution which works by sorting the integers. I am still not sure this is the most efficient way to solve the problem but at least I was able to verify it works properly this way.